Al Rabeeh Dental Center • EMRS Architecture & Workflow Blueprint

Executive technical specification mapping the 11-module clinical reception sequence, 4-tier microservices architecture, Bahrain CPR card reader driver integration, RabbitMQ asynchronous queuing, and offline-first zero-data-loss synchronization.

REACTJS + PHP 8.3 MVC | .NET SMART CPR | OFFLINE AUTO-SYNC

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11-Module Unified Navigation Sequence (CORE-02 Architecture)

IDENTICAL SEQUENCE FOR RECEPTION • DOCTOR • ADMIN

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Every user role navigates through the same standardized 11-module clinical sequence. From reception check-in through dental charting, treatment recalls, lab coordination, and accounts closure, data flows seamlessly without manual re-entry:

01

Dashboard & 6 KPIs

Real-time collections, call volumes, reviews, and cross-department referral metrics.

Front-Desk Cockpit

02

Calendar & Smart CPR

Multi-doctor drag-and-drop scheduler, 20 status counters, and 1-scan Bahrain CPR card reader.

Patient Check-In

03

MSD Leakage Tracker

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Missing Schedule Day tracker automatically captures unbooked patients across 7 cohorts.

Safety Net #1

04

Fail Visit Tracker

Yesterday unattended no-show rollover with 10 call outcome filters and deduplication locks.

Safety Net #2

05

Ortho Continuity

Doctor running patient lists, 5 appliance categories, and 1-6+ month overdue aging buckets.

Safety Net #3

06

Treatment Plan (TP)

3-rule active filter, BHD value metrics, Favorite TP cohort, and FDI tooth-level clinical recall.

Safety Net #4

07

Feedback Calls

Automated post-treatment triggers, live percentage funnel, and dual doctor/reception notes.

Patient Retention

08

Dental Lab Workflow

3-zone shade master, 3-stage delivery pipeline, overdue alerts, and 3-lane internal/external chat.

Lab Coordination

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09

Insurance Claims

Globemed & Mednet submission formats, deductible calculation engine, and 6 attachments.

Claims Engine

10

Patient Journey

24-column clinical compliance matrix, chronological seen audits, and tablet e-consent signatures.

Quality Assurance

11

Billing & Accounts

Cashier multi-payment ledger (Cash, BenefitPay, Credimax), collection auto-sync, and shift sign-off.

Financial Closure

Enterprise Technology Stack Architecture (Tiered Visual Model)

ZERO-TRUST • OFFLINE RESILIENT

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Tier 1: Presentation & Front-Office Edge

CLIENT TIER

ReactJS 18 SPA

Component-driven front-office UI with responsive layout, real-time status counters, and instant keyboard shortcuts.

.NET Smart CPR Service

Local Windows driver communicating over USB smart card reader to extract CPR demographics directly into booking.

Touch POS & Tablet E-Consent

Digital signature pads for patient medical history consents, procedure approvals, and photo link attachments.

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Encrypted Local Bus • Dual-Write Strategy

Tier 2: Edge Persistence & Offline Shield

RESILIENCE TIER

SQLite / IndexedDB Local Storage

High-concurrency WAL local database caches clinic schedule and patient records for continuous offline operations.

AES-256-GCM Cryptographic Vault

Zero-trust cryptographic isolation encrypts CPR card numbers, medical history, and clinical diagnosis records at rest.

Offline Staging Dispatcher

Maintains transactional outbox queue when internet drops; replays staged transactions sequentially upon reconnect.

AMQP 0-9-1 / TLS 1.3 Asynchronous Pipeline

Tier 3: Distributed Middleware & Message Queue

BROKER TIER

RabbitMQ Message Broker

Dedicated queues decouple front-desk UI from heavy background jobs (recall sweeps, automated sync, audit logging).

WhatsApp & SMS Gateway

Automated appointment reminders, calendar confirmations, and post-procedure recall notices dispatched via API.

Sync Daemon & Dead-Letter Queue

Automated exponential backoff retries guarantee 100% data reconciliation across all 6 clinics without duplicates.

FastCGI / Prepared Statements (PDO)

Tier 4: Enterprise Backend & Multi-Branch Database

CORE TIER

PHP 8.3 MVC Microservices

High-performance REST API services for appointment scheduling, treatment plans, lab coordination, and billing ledger.

MySQL 8.0+ Multi-Branch DB

Enterprise relational database with InnoDB foreign keys, strict ACID compliance, and multi-branch replication.

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Bahrain Payment Gateways

Unified checkout integration supporting BenefitPay QR, Credimax, and EazyPay with automated day-closing reconciliation.

11 Modules Comprehensive Functional Breakdown

100% BLUEPRINT ALIGNED

Mod 1: Dashboard & 6 KPIs

Module 01

- Calls, Collections, IG Reviews, Google Reviews cards
- Dental-to-Ortho & Ortho-to-Dental referral funnel
- 24-Hour live audit trail and doctor target tracking

Mod 2: Multi-Doctor Calendar

Module 02

- Drag-and-drop multi-doctor appointment grid
- 1-Scan Bahrain CPR smart card reader (.NET Driver)
- 20 live status filter counters & Bulk WhatsApp API

Mod 3: MSD Leakage Tracker

Module 03

- Missing Schedule Day automated detection
- 7 clinical cohort cards (RCT, Scaling, Crown, etc.)
- 1-click booking handoff with instant unbooked removal

Mod 4: Fail Visit Recovery

Module 04

- Automated yesterday unattended rollover engine
- 10 call outcome filters (No Answer, Rescheduled, Busy)
- Interactive call history log modal & deduplication lock

Mod 5: Ortho Continuity Tracker

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- Doctor running patient list (active monthly cases)

5 appliance classification categories

7 overdue aging buckets (1, 2, 3, 4, 5, 6+ Months)

Mod 6: Treatment Plan (TP)

Module 06

3-rule active filter with BHD treatment plan values

Favorite Treatment Plan patient tracking

7 tooth-level clinical recall cohorts (FDI Notation)

Mod 7: Feedback Calls

Module 07

Auto-triggers upon procedure completion

Real-time call percentage funnel analytics

Dual doctor and receptionist resolution lanes

Mod 8: Dental Lab Workflow

Module 08

3-zone visual shade master (Gingival/Body/Incisal)

3-stage pipeline (Sent, Received, Fitted)

3-lane internal/external chat with overdue alerts

Mod 9: Insurance Claims

Module 09

Globemed & Mednet standard export formats

Deductible, co-pay & patient receivable calculator

6 mandatory attachment checks (X-Rays, E-Sign, etc.)

Mod 10: Patient Journey Matrix

Module 10

24-column clinical compliance matrix

Chronological patient seen audit trail

Interactive tablet e-consent signatures & photos

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Mod 11: Billing & Accounts

- Multi-payment cashier ledger (Cash, Card, Benefit)
- Collection auto-sync across all clinical modules
- Formal end-of-shift signed PDF handover report

Offline-First Auto-Sync Architecture (Internet-Droppable Resilience)

ZERO DOWNTIME GUARANTEE

STAGE 1: NORMAL STATE

Live Dual-Write Synchronization

When internet connectivity is active, all reception check-ins, CPR reads, and payments write simultaneously to local IndexedDB/SQLite and stream via WebSocket/REST to the central cloud MySQL database.

STAGE 2: INTERNET DROP

Isolated Local Buffer Mode

If the clinic's ISP drops, front-office operations continue uninterrupted. The system stores bookings, CPR card demographic reads, and payments locally in AES-256 encrypted storage with an append-only transactional log.

STAGE 3: CONNECTION RESTORED

Automated RabbitMQ Replay

The moment internet reconnects, the sync dispatcher automatically flushes staged records into RabbitMQ message queues. An automated reconciliation daemon resolves any timestamp conflicts with zero data loss.

12-Milestone Delivery Roadmap

TURNKEY EXECUTION

MS-01

Core Architecture, Data Models & Reception Dashboard (Module 1) • Weeks 1 - 6 (1.50 Mo)

ReactJS + PHP MVC microservices framework, RabbitMQ se
and Module 1 Reception Dashboard with 6 real-time KPI ca

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MS-02

Multi-Doctor Scheduler, .NET Smart CPR Driver & WhatsApp API (Module 2) • Weeks 7 - 12 (1.50 Mo)

Drag-and-drop doctor calendar, 20 live filter status counters, custom .NET Bahrain CPR smart card reader driver, and bulk WhatsApp/SMS messaging queue.

MS-03 & 04

MSD Leakage Tracker (Mod 3) & Fail Visit Recovery (Mod 4) • Weeks 13 - 19 (2.00 Mo)

Automated unbooked patient engine across 7 cohorts; yesterday unattended no-show rollover with 10 call outcome filters, call history modal, and booking lock.

MS-05 & 06

Ortho Continuity (Mod 5) & Treatment Plan Recall (Mod 6) • Weeks 20 - 28 (2.25 Mo)

Doctor running patient lists, 5 appliance cards, 7 overdue aging buckets; 3-rule active TP filter, BHD value metrics, and FDI tooth-level clinical recall.

MS-07 & 08

Feedback Calls (Mod 7) & Dental Lab Lifecycle (Mod 8) • Weeks 29 - 36 (2.25 Mo)

Procedure completion automated feedback triggers; 3-zone visual shade master, 3-stage delivery pipeline, overdue alerts, and 3-lane internal/external lab chat.

MS-09

Insurance Tracker & Globemed / Mednet Submission (Module 9) • Weeks 37 - 42 (1.50 Mo)

Multi-procedure invoices, deductible/co-pay calculation engine, 6 mandatory attachment verification checkpoints, and Globemed/Mednet PDF & Excel exports.

MS-10 & 11

Patient Journey Compliance (Mod 10) & Cashier Ledger (Mod 11) • Weeks 43 - 50 (2.00 Mo)

24-column clinical compliance matrix, tablet e-consent signatures; receptionist cashier ledger (Cash, Credimax, BenefitPay), collection auto-sync, and shift sign-off.

MS-12

Shared Doctor & Admin Trigger Console + Final Go-Live • Weeks 51 - 56 (1.50 Mo)

Doctor encounter/procedure/tooth/lab trigger console, Admin KPI approval suite, multi-branch pilot testing across 6 clinics, VAPT security audit, and clinical training.

Contract Governance & Intellectual Property Ownership

100% CLIENT

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100% Client Source Code Ownership

Al Rabeeh Dental Center owns all custom ReactJS source code, PHP MVC microservices, MySQL schemas, .NET CPR driver services, and APIs upon final sign-off.

Max 3 Revisions Per Deliverable

Structured review cycles guarantee rapid progress. Any change request beyond 3 iterations generates an agreed 10% milestone adjustment.

Server Hosting at Actuals

Production server hosting provided by client or billed at actual vendor rates. Milestone invoices payable within 7 working days of confirmation.

60-Day Warranty & NHRA PDPL

Complimentary hypercare technical support post-Go-Live with strict Kingdom of Bahrain Personal Data Protection Law (PDPL) compliance.

Al Rabeeh Dental Center × SaNDS Lab Middle East W.L.L.

Turnkey EMRS Architectural & Workflow Specification • Kingdom of Bahrain • ISO 9001:2015 Certified Division

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